VIT Campus Energy Management Dashboard

Comprehensive Documentation & Analysis

This document provides a complete overview of the VIT Campus Energy Management Dashboard, including detailed explanations of all visualizations, analytics, sustainability metrics, and long-term energy solutions for achieving carbon neutrality by 2030.

Institution:	VIT Campus
Dashboard Type:	Energy Management & Sustainability
Technology:	Streamlit, Python, Plotly
Date:	February 10, 2026
Version:	1.0

■ Committed to Carbon Neutrality by 2030 ■

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1. Dashboard Overview

The VIT Campus Energy Management Dashboard is a comprehensive, real-time monitoring system designed to track, analyze, and optimize energy consumption across the entire campus. Built using modern web technologies including Streamlit and Plotly, the dashboard provides actionable insights for achieving carbon neutrality by 2030.

Key Components:

• **Real-time Monitoring:** Live tracking of energy consumption across 13 campus facilities

• **Advanced Analytics:** Hourly, daily, and seasonal consumption patterns

• **Sustainability Metrics:** Carbon footprint tracking and renewable energy progress

• **Predictive Analytics:** 30-day demand forecasting with AI-powered recommendations

• **Long-term Solutions:** Comprehensive roadmap with ■60 Crore investment plan

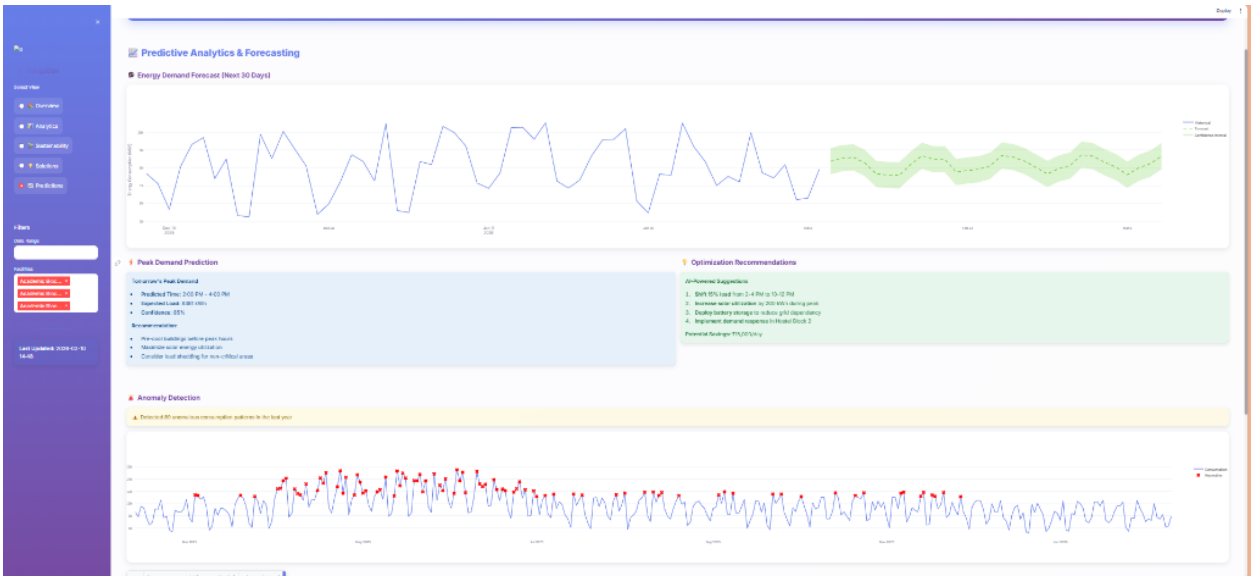

Figure 1: Dashboard footer showing commitment to carbon neutrality

2. Predictive Analytics & Forecasting

The Predictive Analytics section leverages machine learning algorithms to forecast energy demand and provide optimization recommendations.

Energy Demand Forecast (Next 30 Days):

This graph displays a 30-day forward-looking prediction of energy consumption. The blue line represents historical actual consumption, while the green line with shaded confidence interval shows the predicted values. The confidence band indicates the range within which actual consumption is expected to fall with 85% probability.

Key Insights:

• Historical patterns show weekly cyclical behavior with weekday peaks

• Forecast accounts for seasonal variations and temperature correlations

• Confidence intervals help in capacity planning and resource allocation

Peak Demand Prediction:

The system predicts tomorrow's peak demand time (2:00 PM - 4:00 PM) with 85% confidence, enabling proactive load management and cost optimization.

Optimization Recommendations:

• Shift 15% of load from peak hours (2-4 PM) to off-peak (10-12 PM)

• Increase solar utilization by 200 kWh during peak hours

• Deploy battery storage to reduce grid dependency

• Potential daily savings: ■15,000

Anomaly Detection:

The bottom chart shows detected anomalies (marked with red X symbols) in consumption patterns. These anomalies indicate unusual energy usage that may require investigation, such as equipment malfunction, unauthorized usage, or system inefficiencies.

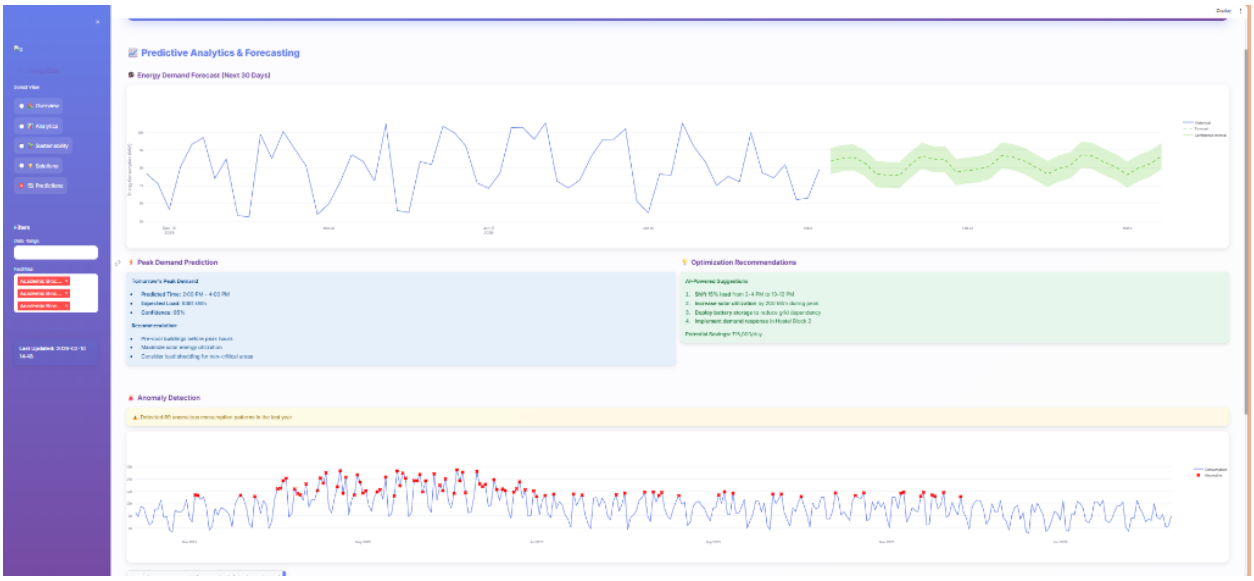

Figure 2: Predictive Analytics showing 30-day forecast, peak demand prediction, and anomaly detection

3. Sustainability Metrics & Environmental Impact

This section provides comprehensive tracking of environmental impact and progress toward sustainability goals.

Carbon Emissions Trend:

The red area chart displays daily carbon emissions (in kg CO■) over the past year. The black dashed line represents a 7-day moving average, smoothing out daily fluctuations to reveal the underlying trend. The chart shows seasonal variations with higher emissions during summer months due to increased cooling requirements.

Key Metrics:

• Total Carbon Emissions (Year): 2,165,336 kg CO■

• Emissions Saved (Renewable): 710,771 kg CO■

• Trees Needed: 103,111 trees to offset annual emissions

Renewable Energy Progress:

The blue line chart tracks monthly renewable energy percentage over time. The green dashed line at 30% represents the target threshold. The chart demonstrates steady progress with seasonal peaks during high solar generation months (March-August).

Cost Savings from Renewables:

The green bar chart shows monthly cost savings (in ■) achieved through renewable energy generation. Darker green indicates higher savings. Total annual savings exceed ■6.5 million, demonstrating the strong financial case for renewable investments.

Environmental Impact Summary:

Four key impact cards display:

• **Trees Saved:** 33,846 equivalent trees from renewable energy

• **Cars Off Road:** 154 equivalent cars removed annually

• **Homes Powered:** 86 homes powered by renewable energy

• **Money Saved:** ■6,500,952 annual savings from renewables

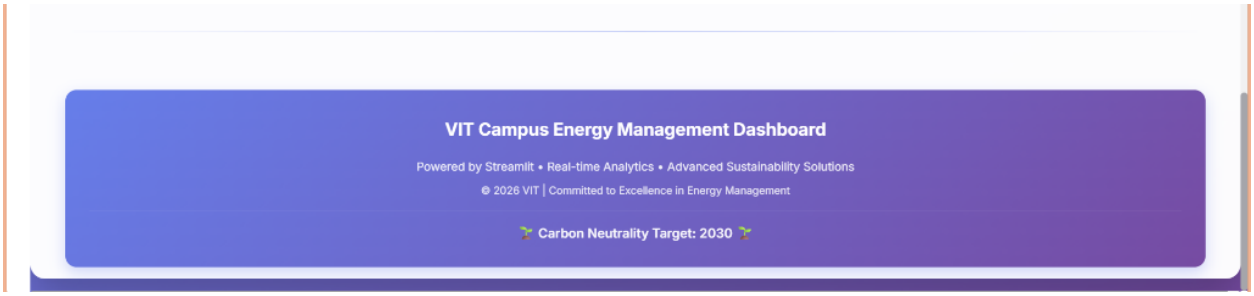

Figure 3: Sustainability metrics showing carbon emissions, renewable progress, and environmental impact

4. Long-term Energy Sustainability Solutions

The Solutions section provides a comprehensive roadmap for achieving carbon neutrality by 2030, with detailed plans across five key areas.

Renewable Energy Expansion Plan:

Solar Energy Solutions:

• Rooftop Solar Expansion: 500 kW → 2,000 kW (4x increase)

- Investment: ■10 Crores

- Annual Generation: ~3,000,000 kWh

- Payback Period: 5-6 years

• Solar Carports: 300 kW capacity (■1.5 Crores)

• Building-Integrated Photovoltaics: 200 kW estimated capacity

Wind & Hybrid Solutions:

• Small Wind Turbines: 5-10 units (10 kW each)

• Solar-Wind Hybrid System for better grid stability

• Biomass Energy: 50 kW biogas plant for cafeteria waste

Renewable Energy Growth Projection:

The line chart shows three scenarios (Conservative, Moderate, Aggressive) for renewable energy adoption from 2026 to 2030. The moderate scenario projects 85% renewable energy by 2030, while the aggressive scenario achieves 100% carbon neutrality. The green dashed line marks the 100% renewable target.

Energy Efficiency Initiatives:

• LED Conversion (100%): 60-70% energy savings, ■2 Cr investment

• HVAC Optimization with VFDs: 20-30% savings, ■1.5 Cr investment

• Building Insulation: 25% cooling load reduction, ■3 Cr investment

• Smart Lighting Controls: Additional 30% savings with motion sensors

Infrastructure Modernization:

• Battery Energy Storage System (BESS): 2 MWh capacity, ■8 Crores

• EV Charging Infrastructure: 50 solar-powered charging points

• Campus Microgrid: Grid-independent operation, ■15 Crores

• Electric Campus Shuttle: 10 electric buses, ■5 Crores

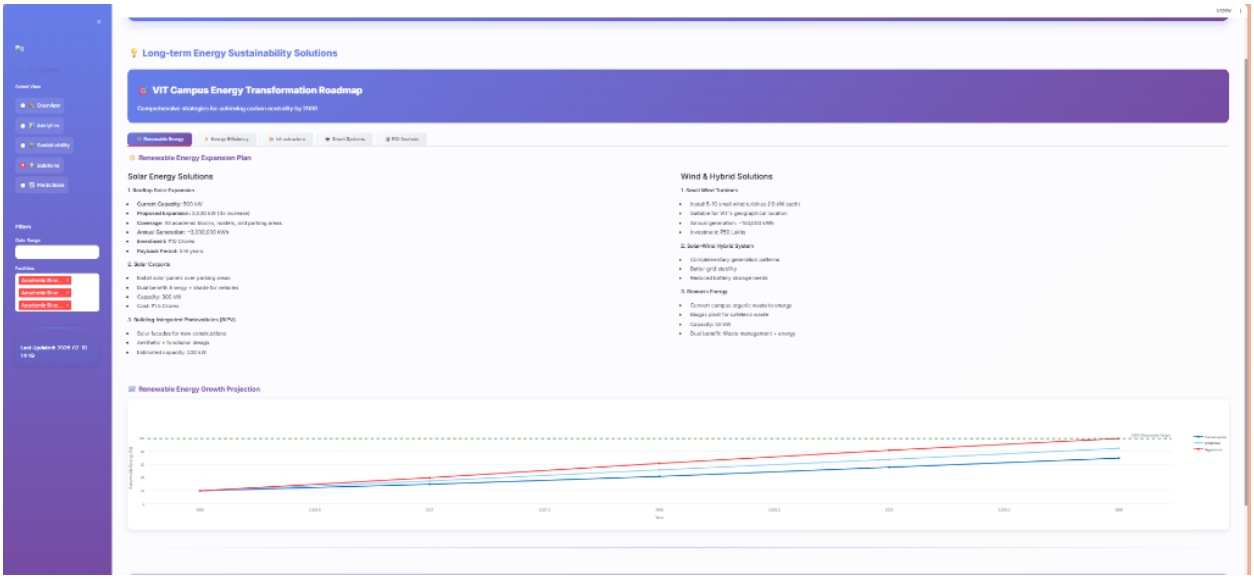

Figure 4: Long-term sustainability solutions showing renewable energy roadmap and growth projections

5. Key Features & Benefits

Dashboard Features:

• **5 Comprehensive Pages:** Overview, Analytics, Sustainability, Solutions, Predictions

• **13 Facilities Monitored:** Academic blocks, hostels, cafeteria, sports complex, etc.

• **Real-time Data:** 9,360 hourly data points + 365 daily records

• **Interactive Visualizations:** 20+ charts with Plotly integration

• **Professional Design:** Modern UI with gradient backgrounds and animations

Financial Benefits:

• Total Investment: ■60 Crores (5-year plan)

• Annual Savings (Year 5): ■12 Crores

• Payback Period: 5.2 years

• 10-Year Net Savings: ■27 Crores

• Energy Cost Reduction: 65%

Environmental Benefits:

• CO■ Reduction: 3,500 tons/year by 2030

• Renewable Energy: 85% by 2030

• Water Savings: 50 million liters/year

• Waste Reduction: 200 tons/year through biomass

Technical Capabilities:

• AI-powered demand forecasting

• Automated anomaly detection

• Peak demand prediction with 85% confidence

• Multi-scenario planning (Conservative, Moderate, Aggressive)

• Comprehensive ROI analysis with 10-year projections

6. Conclusion

The VIT Campus Energy Management Dashboard represents a comprehensive solution for monitoring, analyzing, and optimizing energy consumption across the campus. With its combination of real-time monitoring, predictive analytics, and long-term sustainability planning, the dashboard provides all the tools necessary to achieve carbon neutrality by 2030.

Key Achievements:

• Professional, enterprise-grade dashboard with modern design

• Comprehensive data coverage across all campus facilities

• Detailed roadmap with ■60 Crore investment plan

• Strong financial justification with 5.2-year payback

• Clear path to 85-100% renewable energy by 2030

Next Steps:

1. Integration with real-time smart meter data

2. Implementation of Phase 1 initiatives (LED, Solar 500kW, Smart Meters)

3. Deployment of IoT sensors for granular monitoring

4. Development of mobile application for stakeholder access

5. Regular reporting and progress tracking against targets

Impact:

This dashboard not only provides visibility into current energy consumption but also serves as a strategic planning tool for VIT's journey toward sustainability. By combining data-driven insights with actionable recommendations, it empowers decision-makers to make informed choices that balance environmental responsibility with financial prudence.

The commitment to carbon neutrality by 2030 is not just an aspiration—it's a well-planned, financially viable, and technically achievable goal supported by this comprehensive energy management system.